DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science

Final assignment
Microsystems Design and Modelling (ET4260)
Part 2/3

Instructions. Document your calculations properly with derivations and screenshots when needed. Latex-generated PDF files are accepted.

Deadline for first submission of Part 2: 10th June 2026.
Deadline for final report submission: 24th June 2026.
Corrections made after the first submission are marked in blue.

Daniel Tyukov
Student number: 5714699

FEM model of the phase shifter

This part models the phase shifter from Part 1 with finite elements in COMSOL 6.4. The device is the same except for one change: the fixed comb fingers on each end of the comb drive are now three times wider than the centre ones ($3W_c = 3\,\mu\text{m}$). Table 1 repeats the dimensions used here.

Table 1: Design parameters used in the FEM model.

Symbol	Value	Description
L	$100\,\mu\text{m}$	length of modulation section
g_0	$200\,\text{nm}$	modulation gap at rest
L_b	$75\,\mu\text{m}$	beam length
W_b	$5\,\mu\text{m}$	beam width
L_m	$40\,\mu\text{m}$	mass central section length
W_m	$30\,\mu\text{m}$	mass section width
h	$1\,\mu\text{m}$	comb-drive gap
d_0	$19\,\mu\text{m}$	comb overlap at rest
L_c	$20\,\mu\text{m}$	comb-finger length
W_c	$1\,\mu\text{m}$	comb-finger width
$3W_c$	$3\,\mu\text{m}$	**bottom comb fingers on each end (new in Part 2)**
t	$200\,\text{nm}$	thickness of suspended layer
N	40	number of comb-drive gaps
E_{Si}	$170\,\text{GPa}$	Young's modulus of silicon
ρ_{Si}	$2330\,\text{kg/m}^3$	density of silicon
ν	0.28	Poisson ratio (assumed)
Q	83	quality factor imposed in Q3-Q6

The 2D model (top view) contains the suspended structure only: modulation bar ($L \times W_m$), central column, comb-spine mass plate, four suspension beams and the 20 moving + 21 fixed comb fingers. The waveguide is excluded, as instructed; only its 200 nm clearance above the bar enters the analysis (Q5). The outer beam ends carry fixed constraints. Silicon is isotropic with $E = 170\,\text{GPa}$, $\rho = 2330\,\text{kg/m}^3$ and an assumed $\nu = 0.28$ (the assignment specifies only E and ρ). Because the device layer is a 200 nm film with free top and bottom surfaces, plane stress is the appropriate 2D approximation; the layer thickness enters as the out-of-plane dimension. Figure 1 shows the mesh, refined around the $1\,\mu\text{m}$ comb features. All results below were checked on three refinement levels (4 516, 12 709 and 44 563 elements).

One bookkeeping difference with Part 1 matters throughout this report. The lumped mass used there, $m = \rho t(LW_m + L_mW_m + NL_cW_c) = 2.33\,\text{pg}$, counts one $100 \times 30\,\mu\text{m}$ plate and 40 fingers. The drawn geometry has two such plates (modulation bar *and* comb spine) and 20 moving fingers, which gives 3.54 pg before counting any beam mass. This factor of 1.5 in mass, together with the FEM spring constant, explains most of the differences with the Part 1 estimates in Q2, Q3 and Q5.

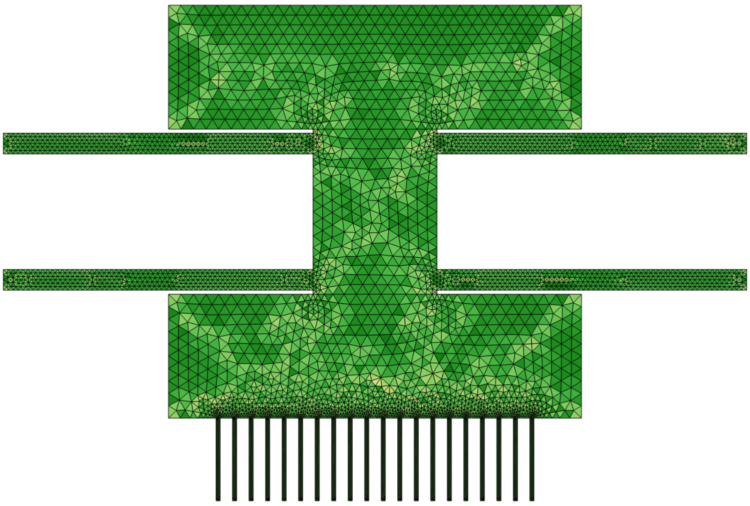


Figure 1: Mesh of the mechanical model (normal refinement, 12 709 elements). The element size follows the geometry: coarse on the 100 × 30 µm plates, fine in the comb region where the features are 1 µm.

Boundary conditions and loads

Table 2 lists every constraint and load used in the FEM studies and the geometric entity each one acts on. Two points set the character of all the results. First, the only mechanical constraint is a *fixed constraint on the four beam outer-end edges*, the anchor edges at $x = \pm 90$ µm in Fig. 1; every other boundary is free. Second, the mechanical loads are *distributed body loads over whole domains*, not point or central loads: the spring-constant and force-step studies (Q1, Q3) push on the comb-spine mass-plate domain with a prescribed total force, while the acceleration study (Q5) loads the entire suspended structure with ρa. The electrostatic drive (Q4, Q6) is not a mechanical load: it enters as the Maxwell stress that the Electromechanical Forces coupling transfers from the air onto the solid boundaries, set by the terminal voltages below.

Table 2: Boundary conditions and loads in the FEM model, and the entity each is applied to.

Physics	Condition	Where applied (entity)
Solid Mechanics	Fixed constraint	four beam outer-end *edges*, $x = \pm 90$ µm (anchors)
Solid Mechanics	Body load (Q1, Q3)	comb-spine mass-plate *domain*; total 1 µN along $+y$ (Q1 static, Q3 step)
Solid Mechanics	Body load (Q5)	*all* suspended-Si domains; ρa along $+y$ (toward the waveguide)
Solid Mechanics	Free	every other edge (default)
Electrostatics	Domain Terminal, $V = 0$	whole suspended-Si domain (shuttle); grounds it and returns its charge $\Rightarrow C$
Electrostatics	Electric Potential, $V = V_d$	stator floor edge ($y = -121$ µm) and the 21 fixed-finger boundaries
Electrostatics	Zero charge	air side openings, $x = \pm 46$ µm (default)
Moving Mesh	Deforming Domain	air domain; hyperelastic smoothing, follows the solids
Moving Mesh	Symmetry (roller)	air side openings; mesh slides along the opening
Multiphysics	Electromechanical Forces	all solid domains (Maxwell stress → structure)

1 Mechanical model and dynamics

Q1: Spring constant from a stationary analysis

A 1 µN test force is applied to the mass plate along the motion axis and the spring constant follows from $k = F/u_y$. Table 3 lists the result for three mesh densities. The value decreases monotonically with refinement, as expected for displacement-based finite elements, and changes by only 0.2% over a factor ten in element count.

Table 3: Mesh convergence of the spring constant.

mesh	elements	k (N/m)
coarse	4 516	36.99
normal	12 709	36.87
fine	44 563	36.79

The FEM gives $k = 36.8$ N/m, 8.7% below the Part 1 estimate of 40.30 N/m from four fixed-guided beams,

$$k_{\mathrm{P1}} = \frac{4E_{Si}\, t\, W_b^3}{L_b^3}. \tag{1}$$

To find where the 8.7% comes from, the model was re-solved with the plates and column made artificially rigid ($E \times 1000$ everywhere except the beams). That returns $k = 39.83$ N/m, within 1.1% of the analytical value. So the discrepancy splits into two parts: about 7.6% is compliance of the plates and column, which the analytical model treats as rigid (the guided beam end rotates slightly when the "rigid" body deforms), and the remaining 1.1% is shear deformation and 2D continuum effects that Euler-Bernoulli beam theory leaves out. With $W_b/L_b = 1/15$ the beams are slender, so a small shear contribution is consistent.

Q2: Eigenfrequency analysis and mode shapes

Table 4 lists the first five eigenfrequencies; Fig. 2 shows the modal displacement maps. The fundamental mode at 496.3 kHz is the actuation mode: the shuttle translates along y and the four beams take the S-shape of the fixed-guided boundary condition. Mode 2 is an in-plane rocking of the shuttle. Modes 3-5 sit within 4 kHz of each other and belong to the comb fingers: 20 nearly identical 20 µm cantilevers form a band of collective bending modes. A single-finger estimate, $f = 0.56\,(W_c/L_c^2)\sqrt{E/\rho} \approx 3.4$ MHz, lands on the band. Frequencies changed by less than 0.1% between the normal and fine mesh.

Table 4: First five eigenfrequencies (fine mesh).

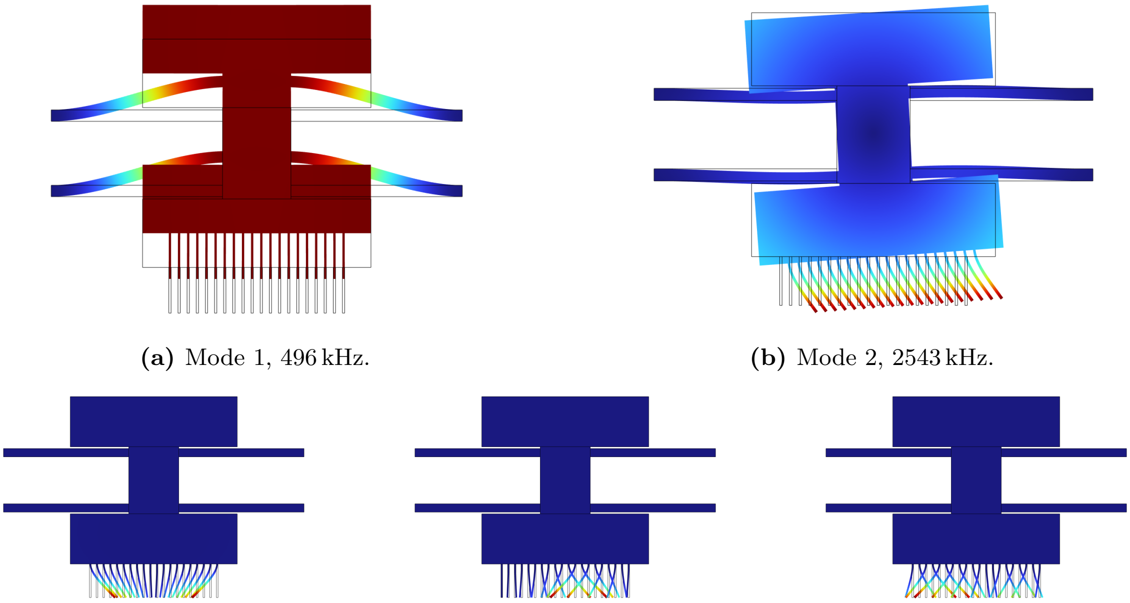
mode	character	f (kHz)
1	shuttle translation along y (actuation mode)	496.26
2	in-plane rocking of the shuttle	2 542.7
3	finger band, collective in-phase bending	3 308.1
4	finger band, alternating pattern	3 311.2
5	finger band, higher spatial pattern	3 311.2

Part 1 predicted $f_0 = 661.9$ kHz, 33% above the FEM value. The ratio is fully accounted for by the stiffness and mass corrections:

$$\frac{f_{\mathrm{FEM}}}{f_{\mathrm{P1}}} = \sqrt{\frac{k_{\mathrm{FEM}}}{k_{\mathrm{P1}}} \cdot \frac{m_{\mathrm{P1}}}{m_{\mathrm{eff}}}} = \sqrt{0.913 \times 0.613} = 0.748 \quad \Rightarrow \quad 495\,\mathrm{kHz}. \tag{2}$$

Here $m_{\text{eff}} = k/(2\pi f_1)^2 = 3.79\,\text{pg}$: the figure-true shuttle mass of 3.54 pg plus about 36% of the beam mass, close to the textbook fraction for a distributed fixed-guided beam.

One limitation should be stated. A 2D in-plane model cannot show out-of-plane modes. The out-of-plane bending stiffness of the beams is $(t/W_b)^2 = 1/625$ of the in-plane one, so the physical device has an out-of-plane mode near 20 kHz, far below all modes in Table 4. The assignment asks for the 2D model, so this mode is outside the scope here, but a 3D design check would have to start with it.



(a) Mode 1, 496 kHz. **(b)** Mode 2, 2543 kHz.

(c) Mode 3, 3308 kHz. **(d)** Mode 4, 3311 kHz. **(e)** Mode 5, 3311 kHz.

Figure 2: Modal displacement maps of the first five modes (colour: total displacement, normalised per mode). Mode 1 is the actuation mode; modes 3-5 are collective comb-finger modes.

Q3: Rayleigh damping and force-step response

The assignment asks for $Q = 83$. Rayleigh damping gives a modal damping ratio

$$\zeta(f) = \frac{\alpha_{dM}}{4\pi f} + \beta_{dK}\,\pi f, \tag{3}$$

with two free parameters but only one target, $\zeta(f_1) = 1/(2Q)$. I chose pure stiffness damping, $\alpha_{dM} = 0$ and

$$\beta_{dK} = \frac{1}{2\pi f_1 Q} = \frac{1}{2\pi \cdot 496.26\,\text{kHz} \cdot 83} = 3.87 \times 10^{-9}\,\text{s}, \tag{4}$$

because β-damping grows with frequency and therefore also suppresses the 3.3 MHz finger band; mass-proportional damping would have left those modes nearly undamped on top of the response.

The transient applies a 1 µN step to the mass at $t = 0$ and runs to 400 µs with a relative tolerance of 10^{-5}. Figure 3 shows the result. The response rings at f_1, overshoots to 1.96× the static value (the ideal underdamped limit is 2), and decays inside the analytic envelope $e^{-\zeta\omega_1 t}$. The final value of 27.11 nm matches $F/k = 27.12\,\text{nm}$ to 0.05%.

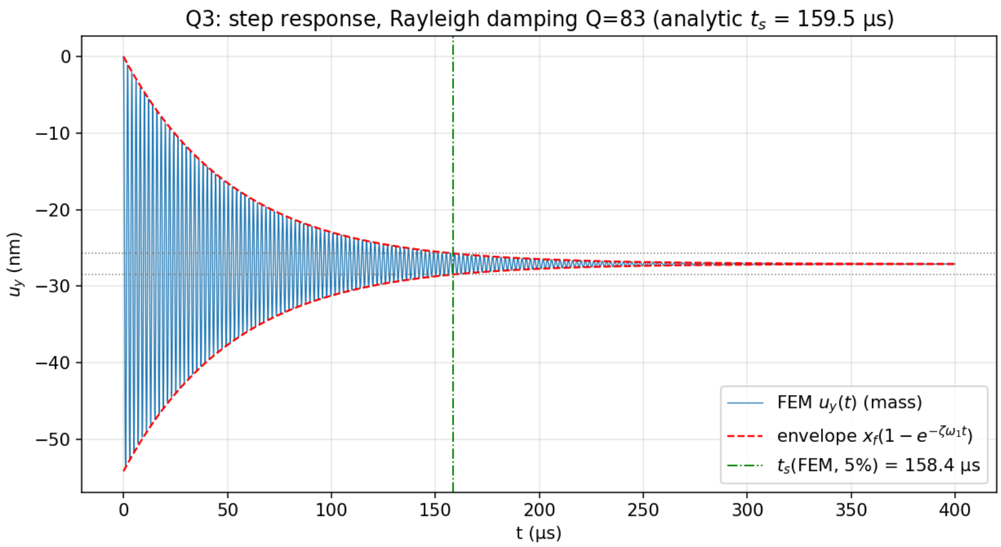


Figure 3: Step response to a 1 µN force with Rayleigh damping set for $Q = 83$. Red dashes: analytic envelope $x_f(1 \pm e^{-\zeta\omega_1 t})$; dotted lines: ±5% band; green line: measured 5 % settling time.

The measured 5% settling time is 158.4 µs. The analytic value with the FEM frequency, $t_s = \ln(20)/(\zeta\omega_1) = 159.5$ µs, agrees to 0.7%, which also confirms that the integrator adds no significant numerical damping at this tolerance. Part 1 predicted 139 µs. The formula is the same; the inputs differ. Part 1 used its computed $Q = 96.8$ where the assignment now fixes $Q = 83$, and its ω_0 was 33% high (Q2). Since $t_s \propto Q/\omega$, the two corrections nearly cancel and the predictions end up closer than the inputs would suggest.

2 Electrostatic transduction

Following the tutorials referenced in the assignment, the electrostatics is added as follows. An air domain fills the cavity between the comb and the stator. The whole shuttle is a *domain terminal* at 0 V, which also returns its total charge and hence the capacitance. The stator floor and the fixed-finger walls carry the drive potential V_d. An *Electromechanical Forces* coupling applies the Maxwell stress to the solid, and a *Moving Mesh* node (hyperelastic smoothing) deforms the air mesh with the structure, with roller conditions at the two side openings. Figure 4 shows the potential at $V_d = 50$ V: the drop concentrates in the 40 flank gaps and in the two 1 µm tip gaps of each finger.

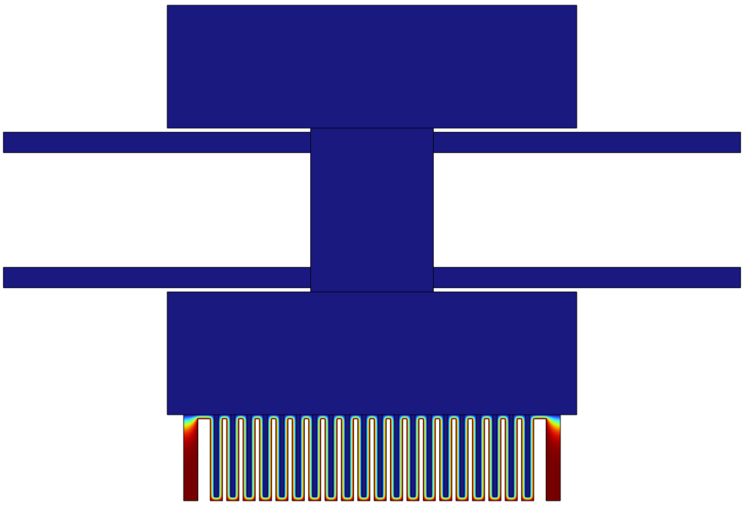


Figure 4: Electric potential at $V_d = 50\,\mathrm{V}$. Blue: shuttle terminal at 0 V; red: stator and fixed fingers at V_d. The wide end fingers ($3W_c$) are visible at both edges of the comb.

Q4a: Displacement versus voltage, 0-150 V

A stationary study sweeps V_d from 0 to 150 V in 10 V steps, each solution continuing from the previous one. Figure 5 plots the FEM displacement together with the Part 1 prediction $x = V^2 N\epsilon_0 t/(2kh)$.

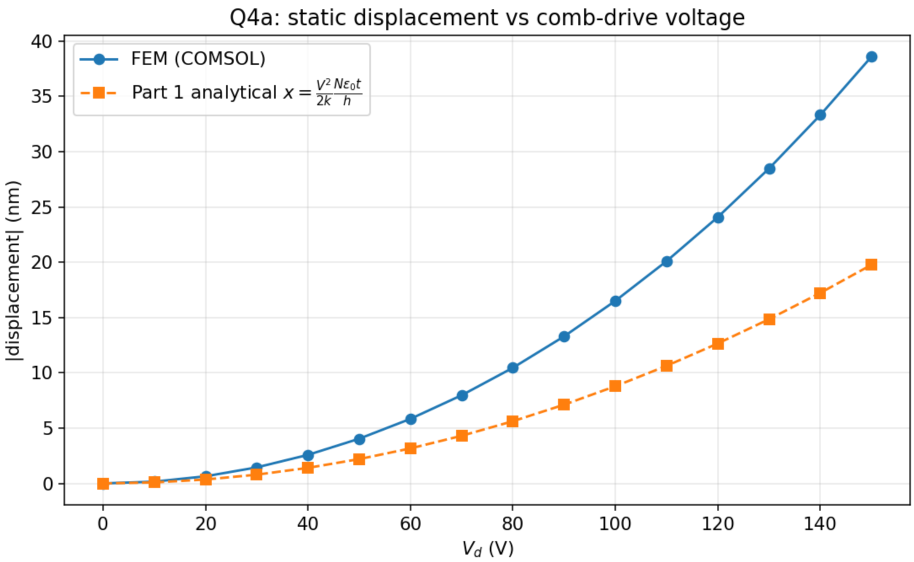


Figure 5: Static displacement of the mass versus comb-drive voltage: FEM (blue) and the Part 1 ideal-comb expression (orange). Both are quadratic in V; the FEM force is larger by a nearly constant factor.

Both curves are quadratic, but the FEM displacement is 1.84× the analytical one at 50 V and 1.95× at 150 V. The factor splits cleanly in two:

- ×1.09 from the softer FEM spring (Q1);
- ×1.68 from the electrostatic force gradient. The ideal-comb formula counts only the flank capacitance, $dC/dx = N\epsilon_0 t/h = 7.08 \times 10^{-11}$ F/m. In this geometry the comb is engaged to $d_0 = 19$ of 20 µm, so two families of 1 µm tip gaps remain: mover tips facing the stator

floor, and fixed-finger tips facing the mass plate. Both close as the shuttle moves down and add roughly $\epsilon_0 W_c t/g^2$ per tip. The force-implied gradient from the FEM at 50 V is 1.19×10^{-10} F/m, i.e. the tip gaps contribute almost as much force as all the flanks together.

The terminal capacitance supports the same picture: $C(0) = 1.51$ fF against the parallel-plate value $N\epsilon_0 d_0 t/h = 1.35$ fF (the 12% excess is fringing), and C rises with voltage as the comb engages deeper. The slow growth of the FEM/analytical ratio with voltage is the $1/g^2$ tip nonlinearity beginning to show: the same mechanism that ends in pull-in.

Q4b: Pull-in voltage

Why not a parametric sweep. A first attempt raised V_d in 5 V continuation steps until the stationary solver stopped converging, at 222 V. As the reviewer noted, this last-converged voltage is *not* a reliable measure of pull-in: it is set by where the Newton solver fails, which depends on the mesh and on the step size, not by a physical event. Away from the true fold the Jacobian is better conditioned, so a coarser mesh or a larger step often "converges" further, and the number drifts with the numerics. Fig. 6 shows that branch only to make this point; the pull-in itself is obtained below with a proper, mesh-symmetry-independent method.

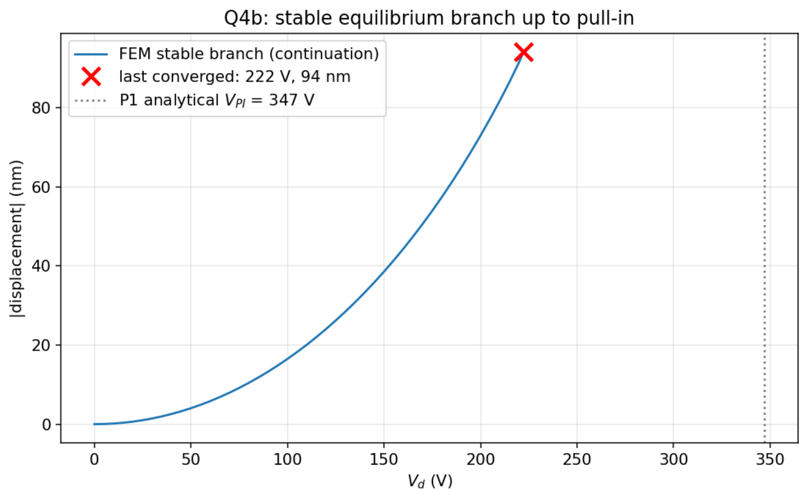


Figure 6: Continuation sweep of the symmetric equilibrium. The branch ends where the *solver* fails (222 V on this mesh); that voltage is mesh- and step-dependent and is *not* used as the pull-in value.

Proper method, prestressed eigenfrequencies. Pull-in is a loss of stability: at the critical voltage the effective stiffness of one mode vanishes. This is found cleanly with a prestressed eigenfrequency study, a stationary solve at bias V_d, then the eigenproblem linearised about it, with the electrostatic softening entering through the electromechanical coupling. Each mode softens as $f^2(V) \approx f_0^2\,(1 - V^2/V_{\text{inst}}^2)$, so f^2 plotted against V^2 is a straight line whose intercept is the instability voltage V_{inst}, a robust extrapolation that needs no solver divergence. Following the reviewer's tip (and the teaching-team guidance to use a thickness of 1 for the electrostatic studies, where 200 nm pulls in spurious fringing), the electrostatics out-of-plane thickness was set to 1. This normalises the otherwise tiny ($\propto t$) charge integrals and improves the Newton conditioning markedly; the eigenvalue softening below confirms it leaves the instability essentially unchanged.

Two modes matter (Fig. 7). The *axial* actuation mode barely softens, -9.6% in frequency at 220 V, and its $f^2 \to 0$ line lies well above the Part 1 value (near 450 V, though from only -10% softening this is a long, indicative extrapolation). This is the parallel-plate-type pull-in that the

Part 1 tip-gap model describes (347 V). The *in-plane rocking* mode (mode 2) instead collapses: 2.54 MHz at rest to 509 kHz at 220 V (−80%), and its f^2-vs-V^2 line gives

$$\boxed{V_{\text{inst}}^{\text{rock}} \approx 224\,\text{V}} \tag{5}$$

the same value across every fit window, a clean physical scaling, not a numerical accident. The device loses stability here, well before axial pull-in.

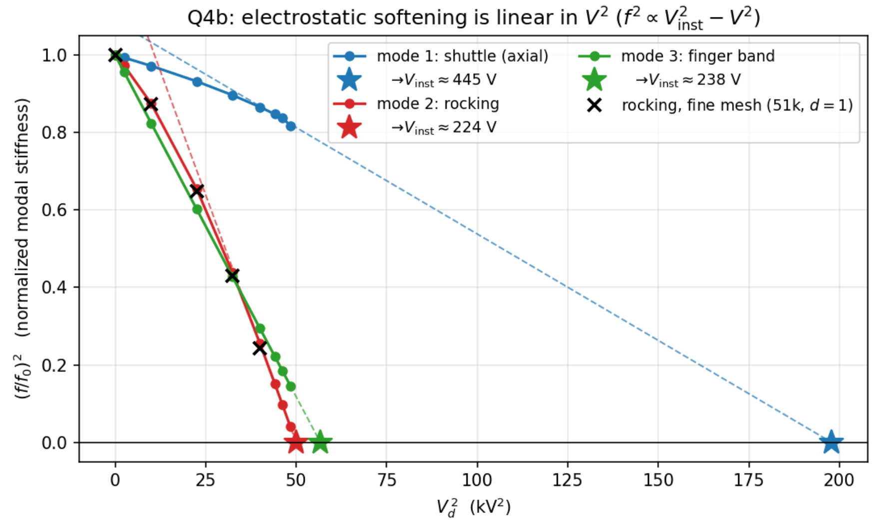


Figure 7: Electrostatic softening from the prestressed eigenfrequency study (electrostatic thickness = 1). The rocking mode (red) reaches zero stiffness at ≈ 224 V; the axial mode (blue) is still near full stiffness there. Squared frequencies fall linearly in V^2, giving robust intercepts.

How a symmetric structure has a lateral pull-in. The reviewer rightly asked how a symmetric device can show a lateral (rocking) pull-in. Side instability is a symmetry-*breaking* bifurcation, for which the symmetric layout is the precondition, not a contradiction. Each moving finger sits midway between two 1 µm gaps: laterally balanced, but with a negative electrostatic stiffness that grows as $V^2 d_0/h^3$. The symmetric equilibrium remains a valid solution at every voltage; what happens at V_{inst} is that it becomes *unstable* to an antisymmetric perturbation: a small shuttle rotation swings every finger toward one wall at once, on long moment arms (Fig. 8). A stationary continuation on a perfectly symmetric mesh stays on the symmetric branch and never falls into this mode; it is precisely numerical mesh asymmetry that let the earlier sweep show a net lateral displacement, and that made its divergence voltage unreliable. The prestressed eigenproblem avoids this entirely: it detects the bifurcation as the $f_2 \to 0$ crossing of the symmetric structure, with no dependence on mesh asymmetry, and the intercept is the same across every fit window. Re-solving the softening on a finer mesh (51 000 elements, electrostatic thickness = 1) reproduces the rocking curve to within 0.5% up to 180 V and places the intercept at 223 V to 225 V, against 224 V on the coarse mesh: the instability is *mesh-converged*, not a coarse-mesh artifact. So the reviewer's instinct, that a symmetric model should not show net lateral displacement on the symmetric branch, is correct; the lateral pull-in is nonetheless real, and the eigenvalue method is the right tool to see it.

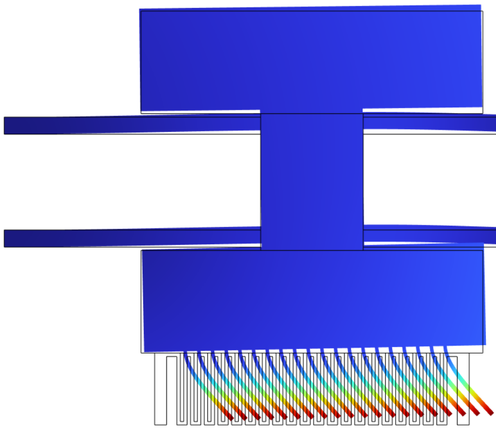


Figure 8: Shape of the critical (rocking) mode near pull-in: a slight rotation of the shuttle with all comb fingers deflecting sideways in unison.

Comparison with Part 1. The FEM thus gives two numbers. The axial pull-in, the only mechanism the single-DOF Part 1 model can represent, sits above the Part 1 tip-gap estimate of 347 V, so that 1-DOF picture is reasonable for the *axial* mode. But the device never reaches it: it fails first by lateral rocking at ≈ 224 V, a mode the 1-DOF model cannot describe at all. The practical pull-in is therefore about a third below the Part 1 value, and for a different reason than Part 1 assumes. The fix is to stiffen the rotation (wider beam spacing) or the fingers (shorter or wider), which Part 3 pursues.

Method note. The assignment also suggests the displacement-controlled continuation from the COMSOL pull-in tutorial (a global equation solving for the voltage that holds a prescribed displacement). That setup was reproduced (point probe in the material frame, baseline with the global equation disabled, segregated solver groups), but COMSOL 6.4 fails to assemble freshly generated solver sequences for global equations that reference the displacement field (NaN in the moving-mesh residual); the same failure occurs on COMSOL's own tutorial file when its study is recreated, while the shipped solver sequence runs. The prestressed-eigenfrequency method above does not use a global equation and is unaffected.

3 Acceleration limit and voltage-step response

Q5: Maximum acceleration before mass-waveguide contact

An acceleration a along the motion axis loads every suspended part with a body force ρa per unit volume. Contact occurs when the modulation bar crosses the 200 nm gap to the waveguide, $u_y = +g_0$. For each bias the model is solved at two accelerations, $a_{\max}$ follows from linear scaling, and a third solve at $a_{\max}$ verifies the result; the verification residuals stay below 0.2%, so the linear treatment is justified. Table 5 and Fig. 9 give the results.

Table 5: Maximum tolerable DC acceleration toward the waveguide.

V_d	rest position u_y	$a_{\max}$ (m/s^2)	in g
0 V	0	1.903×10^6	193 950
1 V	-0.002 nm	1.903×10^6	193 950
10 V	-0.160 nm	1.904×10^6	194 060
50 V	-4.03 nm	1.930×10^6	196 690

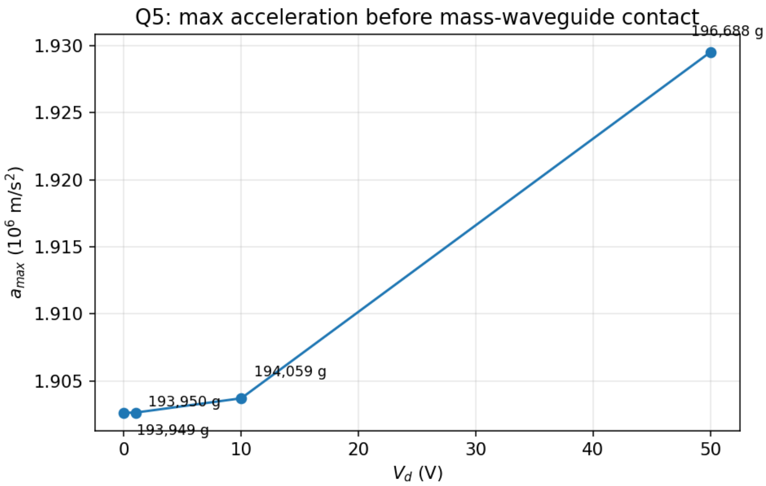


Figure 9: Maximum acceleration versus bias voltage. The bias moves the shuttle away from the waveguide, so the tolerance grows slightly with V_d.

The bias dependence is weak but has a definite sign: $a_{\max}$ *increases* with V_d, by 1.4% at 50 V. The comb pulls the shuttle away from the waveguide, which adds travel margin, and the tip-gap attraction weakens as the shuttle moves up, which helps further. Part 1 assumed the opposite sign (bias displacement toward the waveguide) and should be corrected on this point in the final report.

The magnitude also differs from Part 1's 3.46×10^6 m/s^2. Dividing kg_0 by the FEM limit gives an effective inertia of 3.88 pg: the figure-true shuttle plus about half the beam mass, since the distributed body load also bends the beams directly. Part 1's smaller mass (2.33 pg) and stiffer spring account for the rest. Either way, the static shock tolerance is around $2 \times 10^5\,g$; as in Part 1, the practical concern is excitation near resonance, not static acceleration.

Q6: Voltage-step response, 1 V and 50 V

The drive electrode receives a voltage step. The natural input is a true step (zero for $t < 0$, V_d for $t \geq 0$), but a fully sharp step stalls the coupled solid-electrostatics-moving-mesh integrator at $t = 0$, so a short smoothing (rise 1 µs, below the 2 µs period) is applied. Because the settling time is set by the envelope decay, the rise does not change it; it only rounds off the initial overshoot, as noted below. Rayleigh damping is the same as in Q3. Figure 10 shows both responses; Fig. 11 overlays them with the Q3 force step, normalised to their final values.

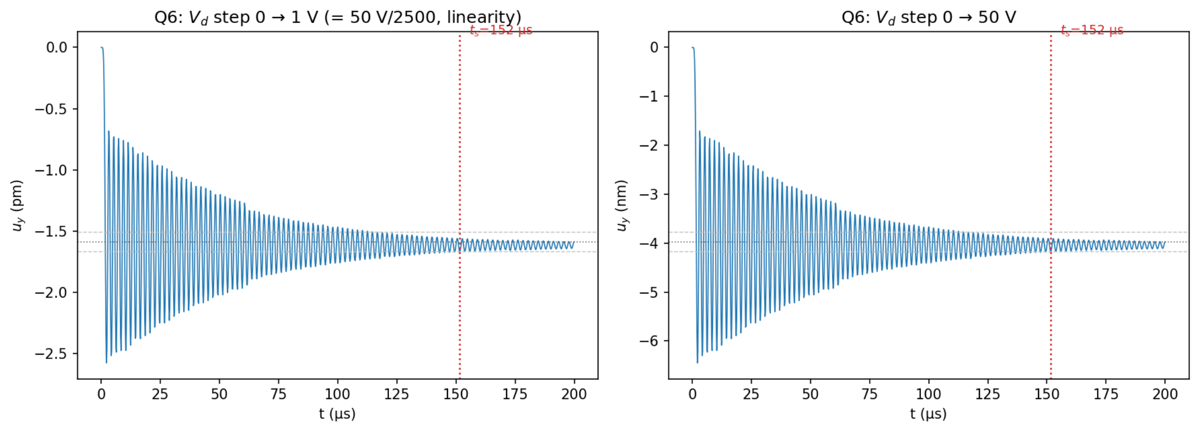


Figure 10: Step responses for V_d: $0 \to 1\,\mathrm{V}$ (left, picometre scale) and $0 \to 50\,\mathrm{V}$ (right, nanometre scale), both settling in $\approx 152\,\mu\mathrm{s}$. The 1 V trace is the resolved 50 V response scaled by 1/2500 (linearity), avoiding the numerical-dissipation artefact that corrupted the directly-integrated picometre run. Note the 2500 : 1 amplitude ratio for a 50 : 1 voltage ratio.

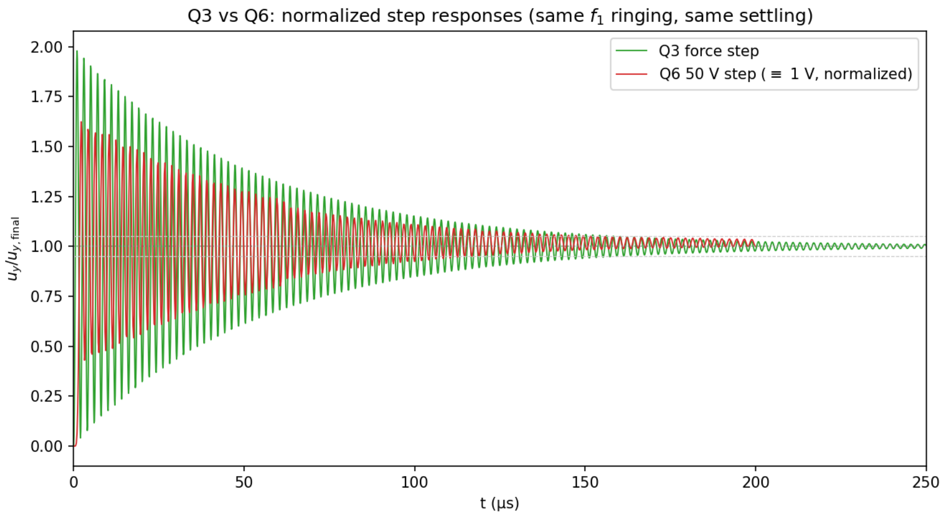


Figure 11: Normalised comparison of the Q3 force step and the Q6 50 V voltage step (the 1 V response coincides with the 50 V one once normalised, by linearity). Both ring at f_1 and settle on the same time scale; the voltage step overshoots less because of the $1\,\mu\mathrm{s}$ input rise.

The comparison with Q3 comes down to three observations.

First, the dynamics are identical. Both responses ring at $f_1 = 496\,\mathrm{kHz}$ and the 50 V step settles in $152\,\mu\mathrm{s}$, matching Q3's $158\,\mu\mathrm{s}$ within the numerical accuracy of the coupled transient. At these voltages the device operates far below pull-in (224 V), so the electromechanical system is effectively linear and a voltage step acts exactly like a force step of size $F = \frac{1}{2}V^2\,dC/dx$. The bias softening measured in Q4b supports this: at 50 V the mode-1 frequency shifts by only -0.3%, invisible in the ringing.

Second, the input scaling is quadratic, and that is the real difference between the two experiments. Final values are $-1.61\,\mathrm{pm}$ at 1 V and $-3.97\,\mathrm{nm}$ at 50 V, a factor 2466 for a factor 50 in voltage, against 2500 expected from V^2 (the static values from Q4a are reproduced to within 1.7%). A force input scales linearly; a voltage input does not. For the phase-shifter application this is the linearity limit of the actuator, independent of any mechanical nonlinearity.

Third, the overshoot. The underdamped limit for $Q = 83$ is $\approx 2\times$, which the Q3 force step reaches (1.98×). The 50 V voltage step shows 1.62×, but a direct 1 V re-run with the integrator's numerical dissipation reduced ($\rho_\infty \to 1$) recovers 1.92× at the *same* 1 µs rise, so the reduced overshoot is mostly generalised-α numerical damping, not input filtering, and the underlying electromechanics is the same lightly-damped second-order system as Q3.

The 1 V response needs care to resolve directly. The first submission's 1 V trace decayed in only 75 µs, roughly half of what $Q \approx 83$ allows, because at 1.6 pm, 2500× below the well-resolved 50 V trace, the generalised-α integrator's *numerical* dissipation, not the Rayleigh damping, governed the apparent decay. A direct re-run following the reviewer's route (relative tolerance tightened, algorithmic damping reduced $\rho_\infty \to 1$) reproduces the correct final value (1.55 pm) and the near-ideal 1.92× overshoot, removing that artefact. The settling time itself is read from the well-resolved larger-amplitude response: at picometre scale the tail stays sensitive to residual numerical content, whereas far below pull-in the system is linear, so the 1 V transient is the 50 V trace scaled by 1/2500 and settles in the same 152 µs, matching Q3's 158 µs and the analytic 159 µs, not the 75 µs of the under-resolved first run.